UCS503P - Software Engineering Project

Project Team Formation Portal

A Web-Based Platform for Project and Hackathon Team Formation

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Contents

Chapter 1

Introduction

1.1 Background and Context

Academic project team formation is an integral component of modern engineering education, specifically in project-based learning courses. Currently, students and faculty rely heavily on informal channels such as messaging groups and word-of-mouth communication to discover projects and teammates. This process lacks a centralized platform for project and hackathon opportunities.

The **Project Team Formation Portal** is a web platform designed to organize project and hackathon opportunities. It provides role-based access, student profiles, and a feed where teachers can publish opportunities with required skills.

1.1.1 Problem Statement

College students face significant hurdles in forming balanced, effective project teams. The existing mechanisms present several key operational challenges:

- **Fragmentation:** Information regarding available projects and open student roles is scattered across informal, unorganized channels.
- **Poor Skill Visibility:** Lack of a centralized student profile containing skills, CGPA, department, and portfolio links.
- **Unstructured Recruitment:** Students currently have limited centralized support for discovering opportunities and forming teams.
- **Limited Team Formation Support:** Team joining, invitations, and team management are planned for future versions.
- **Limited Academic Supervision:** Teachers need a centralized way to publish project and hackathon opportunities for students.

1.2 Project Objectives

The primary goal of this project is to implement a web-based solution that centralizes project and hackathon opportunities and provides a foundation for project team formation.

1. **Centralized Discovery:** Provide students with a feed for viewing project and hackathon opportunities posted by teachers.
2. **Profile Management:** Allow students and teachers to maintain role-specific profiles containing relevant academic, skill, and professional information.
3. **Role-Based Access:** Create dedicated dashboards for Students and Teachers with appropriate access to available features.
4. **Future Team Formation:** Provide a foundation for team joining, invitations, notifications, and calendar features planned for future versions.

Chapter 2

Methodology and Design

2.1 System Architecture

The system architecture follows a decoupled client-server model built around RESTful API communications, persistent storage via MongoDB, and role-based client routing.

2.1.1 High-Level Design

The high-level architecture comprises three major tiers:

1. **Presentation Tier:** A responsive Single Page Application (SPA) built using React.js for seamless user interactions across multi-device endpoints.
2. **Application Tier:** A Node.js and Express.js backend handling API routing, business logic execution, authentication, and authorization checks.
3. **Data Tier:** A MongoDB database storing user profiles and project or hackathon post data.

2.1.2 UML Diagrams and System Workflows

To formalize system functionality and actor interactions, the following UML diagrams describe the main system roles and available workflows.

Use Case Diagram

Figure **??** illustrates student and teacher interactions with the core platform features.

Core Recruitment Sequence

The main interaction sequence for viewing opportunities and publishing a hackathon post is detailed below:

1. **Feed Access:** Student opens the dashboard feed to view available project and hackathon posts.
2. **Profile Information:** Student profile displays department, CGPA, skills, and optional portfolio links.

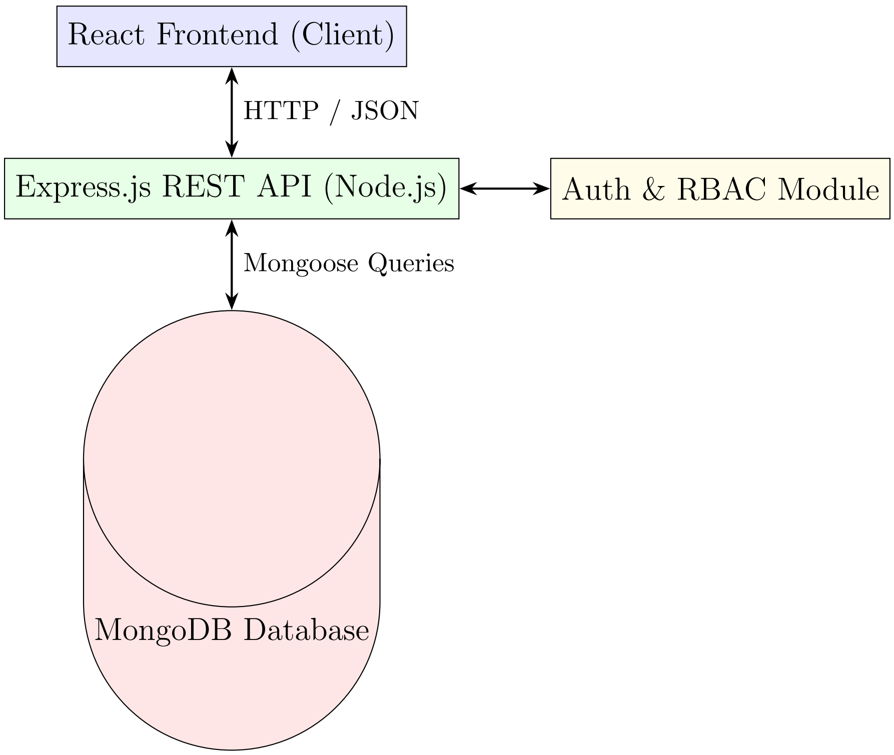

Figure 2.1: High-Level System Architecture Diagram

3. **Team Formation:** Team creation is currently marked as a future feature.

4. **Post Creation:** Teacher enters the hackathon title, description, and required skills.

5. **Feed Update:** The published post becomes available in the student and teacher feeds.

2.1.3 Technical Stack

- **Framework/Language:** JavaScript (ES6+), React.js, Node.js.
- **Backend API Services:** Express.js framework.
- **Database:** MongoDB database with Mongoose.
- **Authentication/Security:** JSON Web Tokens (JWT), bcrypt password hashing.
- **DevOps & Tools:** Git and GitHub.

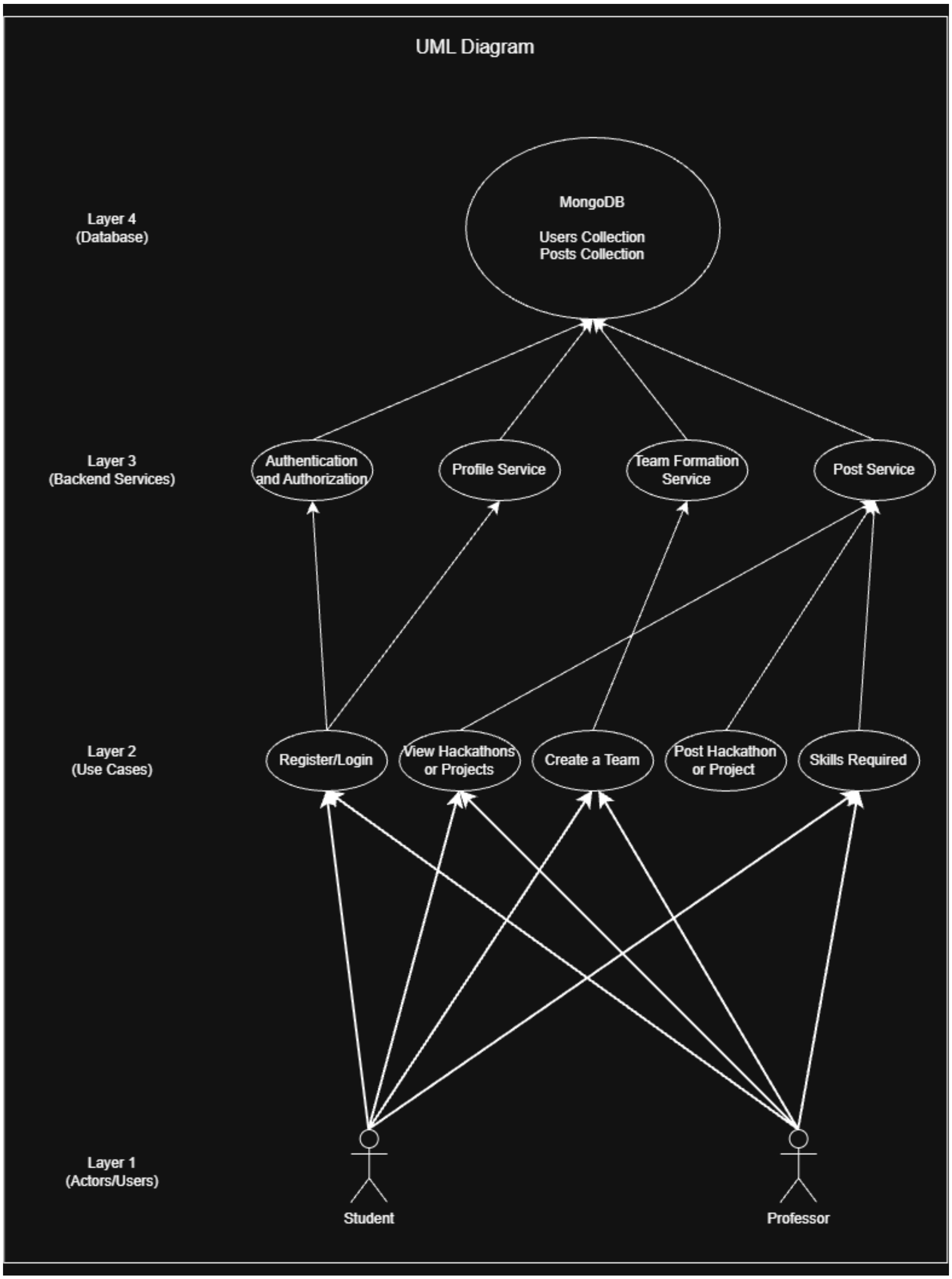

Figure 2.2: Use Case Diagram

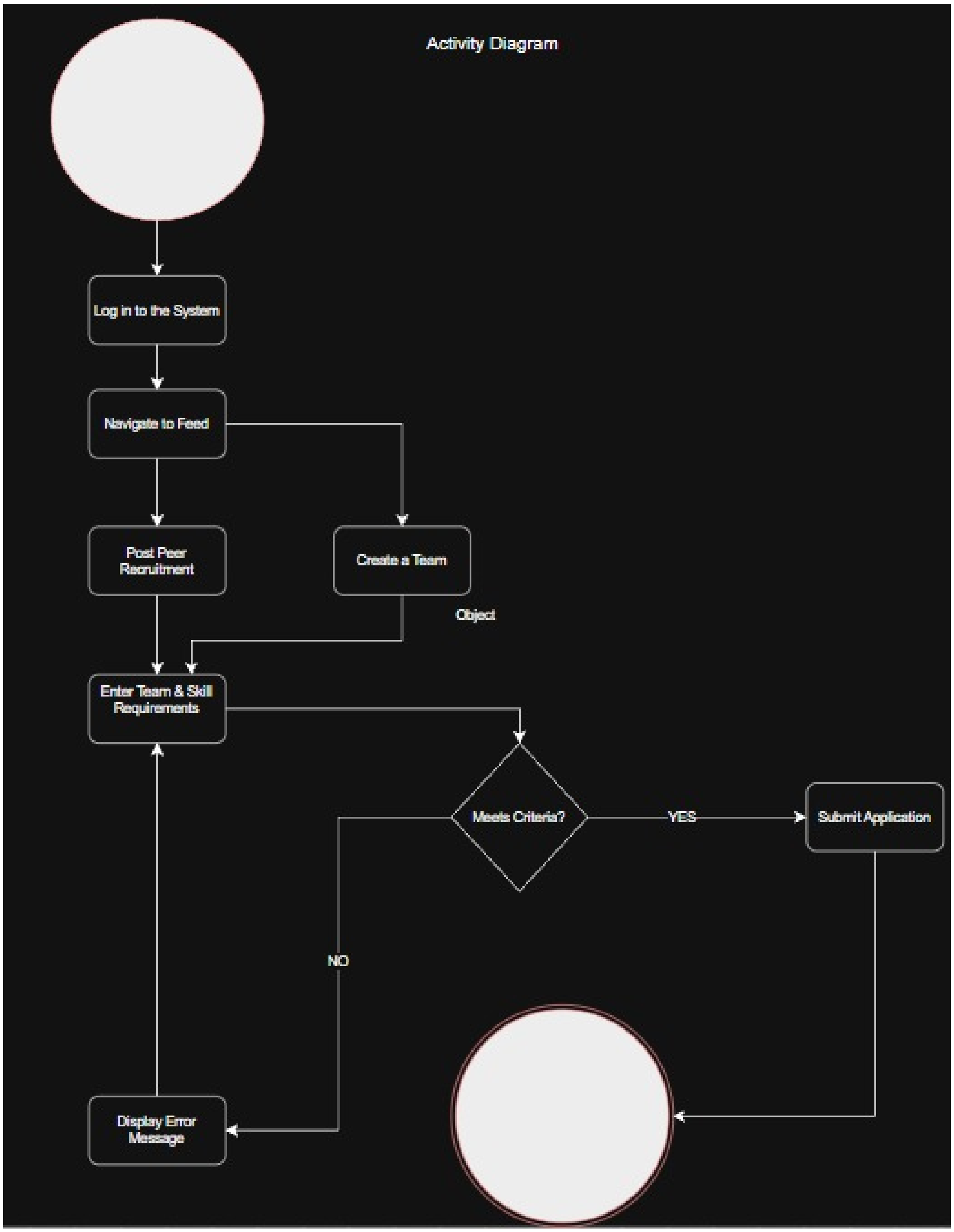

Figure 2.3: Activity Diagram

2.2 Implementation Details

2.2.1 Module 1: User Management & Profile System

Enforces role-based authorization for *Student* and *Teacher* roles. Students construct profiles featuring academic information, CGPA, skills, and optional links such as

GitHub, LinkedIn, and resume information.

2.2.2 Module 2: Project Marketplace & Eligibility Engine

Allows teachers to publish project and hackathon posts specifying a title, description, and required skills. These posts are displayed in the feed.

2.2.3 Module 3: Skill Verification Engine

The skill verification and testing functionality is planned for a future version.

2.2.4 Module 4: Collaboration & Administration Dashboards

Provides separate student and teacher dashboards, with team joining, invitations, notifications, and calendar functionality marked as coming soon.

Chapter 3

Results and Evaluation

3.1 Testing Methodology

A basic functional testing strategy was used to verify the implemented features:

- **Authentication Testing:** Tested registration, login, JWT authentication, and role-based access.
- **Profile Testing:** Tested student and teacher profile creation and profile updates.
- **Post Testing:** Tested teacher hackathon post creation and display of posts in the feed.
- **User Testing:** Tested the student and teacher dashboards, navigation, and Coming Soon features.

3.2 Performance Metrics

Implemented features were tested for correct navigation, authentication, profile handling, and post display.

Feature	**Status**
User Registration and Login	Implemented
Student and Teacher Profiles	Implemented
Teacher Hackathon Posts	Implemented
Feed	Implemented
Team, Notification and Calendar Features	Coming Soon

Table 3.1: Implemented Feature Status

3.3 Analysis

The implemented system successfully provides centralized authentication, role-based dashboards, profile management, and teacher-created hackathon posts that are visible in the feed.

Chapter 4

Conclusion

4.1 Summary of Work

The **Project Team Formation Portal** provides a structured web-based solution for organizing project and hackathon opportunities in academic environments. The platform offers role-based authentication, student and teacher profiles, teacher-created posts, and a centralized feed.

4.2 Key Findings

- **Centralized Opportunities:** Students can view project and hackathon opportunities from a common feed.
- **Role-Based Access:** Students and teachers receive dashboards appropriate to their roles.
- **Simple Recruitment Foundation:** The portal provides a foundation for future team formation and collaboration features.

4.3 Future Work and Recommendations

Planned enhancements for future iterations include:

1. Team creation, joining, and member invitation functionality.
2. Notifications and calendar functionality.
3. Skill verification tests and advanced teammate matching.

4.4 Final Remarks

By converting scattered project and hackathon information into a centralized digital platform, this system establishes a practical foundation for project-based team formation and future collaboration features.